

CSS Text Formatting Properties

Complete Guide: Styling and Manipulating Text

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1 Introduction

CSS text formatting properties control how text appears and behaves on web pages. These properties enable developers to style text for readability, visual emphasis, and better user experience.

1.1 Text Formatting Categories

Text formatting properties fall into several categories:

- **Text Alignment:** How text aligns within containers
- **Text Decoration:** Underlines, overlines, strikethrough
- **Text Transform:** Uppercase, lowercase, capitalization
- **Text Spacing:** Letter and word spacing
- **Text Indentation:** First-line indentation
- **Text Shadow:** Shadow effects on text
- **Text Overflow:** Handling text that doesn't fit
- **Whitespace:** How whitespace is handled
- **Word Breaking:** Text wrapping and breaking

2 Text Alignment

The `text-align` property controls horizontal text alignment.

2.1 `text-align` Values

Value	Behavior	Use
<code>left</code>	Aligns left	Default for LTR text
<code>right</code>	Aligns right	Default for RTL text
<code>center</code>	Centers text	Headings, callouts
<code>justify</code>	Stretches text	Professional documents
<code>start</code>	Flow-relative start	RTL/LTR agnostic
<code>end</code>	Flow-relative end	RTL/LTR agnostic

2.2 `text-align` Examples

```
/* Left aligned (default) */
p {
  text-align: left;
}

/* Centered text */
h1 {
  text-align: center;
}

/* Right aligned */
.right-text {
  text-align: right;
}

/* Justified text */
article {
  text-align: justify;
}

/* Flow-relative (supports RTL) */
.content {
  text-align: start; /* Left in LTR, right in RTL */
}
```

2.3 Vertical Alignment

The `vertical-align` property controls vertical alignment in inline/inline-block contexts:

```
/* Inline elements */
img {
  vertical-align: middle; /* Aligns with text middle */
}

/* Table cells */
td {
```

```
vertical-align: top;
vertical-align: middle;
vertical-align: bottom;
}

/* Inline-block elements */
.icon {
  display: inline-block;
  vertical-align: middle;
}
```

3 Text Decoration

The `text-decoration` property adds decorative lines to text.

3.1 text-decoration Properties

Property	Purpose	Values
<code>text-decoration-line</code>	Type of line	underline, overline, line-through
<code>text-decoration-color</code>	Color of line	Any CSS color
<code>text-decoration-style</code>	Line style	solid, dotted, dashed, wavy
<code>text-decoration-thickness</code>	Line thickness	Measurements

3.2 text-decoration Examples

```
/* Underline */
a {
  text-decoration: underline;
}

/* Remove underline */
a {
  text-decoration: none;
}

/* Overline */
.overlined {
  text-decoration: overline;
}

/* Strikethrough */
.strikethrough {
  text-decoration: line-through;
}

/* Multiple decorations */
.decorated {
  text-decoration: underline overline;
}

/* Custom decoration styling */
```

```
a {
  text-decoration-line: underline;
  text-decoration-color: red;
  text-decoration-style: wavy;
  text-decoration-thickness: 2px;
}

/* Shorthand */
a:hover {
  text-decoration: underline wavy red 2px;
}
```

3.3 text-decoration-skip

Control what parts are decorated:

```
a {
  text-decoration: underline;
  text-decoration-skip-ink: auto; /* Skip descenders */
}

/* Values:
  - auto: Skip descenders (default)
  - none: Don't skip anything
  - spaces: Skip spaces
  - leading-spaces: Skip spaces at start
  - trailing-spaces: Skip spaces at end
*/
```

4 Text Transform

The `text-transform` property changes text case.

4.1 text-transform Values

Value	Effect	Example
<code>none</code>	No transformation	(unchanged)
<code>uppercase</code>	All capitals	HELLO WORLD
<code>lowercase</code>	All lowercase	hello world
<code>capitalize</code>	Capitalize first letter	Hello World
<code>full-width</code>	Full-width characters	Used for CJK

4.2 text-transform Examples

```
/* Uppercase */
h1 {
  text-transform: uppercase;
}

/* Lowercase */
.slug {
  text-transform: lowercase;
}

/* Capitalize */
.title {
  text-transform: capitalize;
}

/* Revert transformation */
.normal {
  text-transform: none;
}

/* All caps with letter spacing */
.screaming {
  text-transform: uppercase;
  letter-spacing: 2px;
}

/* Badge styling */
.badge {
  text-transform: uppercase;
  font-weight: 700;
  font-size: 0.75rem;
  letter-spacing: 1px;
}
```

5 Text Spacing

Fine-tune spacing between characters and words (covered partially in Font guide, but revisited here).

5.1 Advanced Spacing Examples

```
/* Increased character spacing */
h1 {
  letter-spacing: 3px;
}

/* Em-based letter spacing */
.elegant {
  letter-spacing: 0.1em;
}

/* Negative letter spacing */
.tight {
  letter-spacing: -1px;
}

/* Word spacing */
p {
  word-spacing: 5px;
}

/* Text indent */
p {
  text-indent: 2em; /* First line indented */
}

/* Remove indent */
p.no-indent {
  text-indent: 0;
}

/* Hanging indent */
ul li {
  text-indent: -1em;
  margin-left: 1em;
}
```

6 Text Shadow

The `text-shadow` property adds shadow effects to text.

6.1 text-shadow Syntax

```
text-shadow: offset-x offset-y blur-radius color;
```


6.2 text-shadow Examples

```

/* Simple shadow */
h1 {
  text-shadow: 2px 2px 4px rgba(0, 0, 0, 0.3);
}

/* Multiple shadows */
h1 {
  text-shadow:
    2px 2px 4px rgba(0, 0, 0, 0.3),
    -2px -2px 4px rgba(255, 255, 255, 0.5);
}

/* Glowing text */
.glow {
  text-shadow:
    0 0 10px #3498db,
    0 0 20px #2980b9;
}

/* Text outline effect */
.outlined {
  text-shadow:
    -1px -1px 0 #000,
    1px -1px 0 #000,
    -1px 1px 0 #000,
    1px 1px 0 #000;
}

/* Embossed effect */
.embossed {
  text-shadow:
    2px 2px 1px rgba(255, 255, 255, 0.8),
    -2px -2px 1px rgba(0, 0, 0, 0.8);
}

```

7 Text Overflow and Wrapping

Handle text that exceeds container boundaries.

7.1 white-space Property

Control whitespace handling:

```

/* Normal wrapping */
p {
  white-space: normal; /* Default */
}

/* No wrapping */
p {

```

```
    white-space: nowrap; /* Single line */
}

/* Preserve whitespace */
pre {
    white-space: pre; /* Like <pre> tag */
}

/* Wrap but preserve whitespace */
code {
    white-space: pre-wrap;
}

/* Collapse whitespace and wrap */
p {
    white-space: normal;
}
```

7.2 text-overflow Property

Show ellipsis when text overflows:

```
/* Single line truncation */
.truncated {
    width: 200px;
    white-space: nowrap;
    overflow: hidden;
    text-overflow: ellipsis;
}

/* Values:
   - clip: Cut off text (default)
   - ellipsis: Show ... at end
   - custom string: Custom overflow character
*/

/* Multi-line truncation */
.truncated-lines {
    display: -webkit-box;
    -webkit-line-clamp: 3; /* Max 3 lines */
    -webkit-box-orient: vertical;
    overflow: hidden;
}
```

7.3 word-break and overflow-wrap

Control how words break:

```
/* Break long words */
.break-word {
    word-break: break-word;
}
```

```
/* Allow breaking at any character */
.break-all {
  word-break: break-all;
}

/* Keep words together */
.keep-all {
  word-break: keep-all; /* For CJK text */
}

/* Wrap instead of breaking */
.overflow-wrap {
  overflow-wrap: break-word;
}

/* Anywhere in word */
.break-anywhere {
  word-break: break-word;
  overflow-wrap: break-word;
}
```

8 Direction and Language

8.1 direction Property

```
/* Left-to-right (default) */
body {
  direction: ltr;
}

/* Right-to-left (for Arabic, Hebrew) */
.arabic-text {
  direction: rtl;
  text-align: right;
}

/* Inherited direction */
.inherited {
  direction: inherit;
}
```

8.2 hyphens Property

Control hyphenation:

```
/* No hyphenation */
.no-hyphens {
  hyphens: none;
}

/* Manual hyphenation */
.manual {
  hyphens: manual; /* Break on soft hyphen (\u00ad) */
}

/* Automatic hyphenation */
.auto-hyphen {
  hyphens: auto;
  lang: en; /* Language needed */
}
```

9 Real-World Examples

9.1 Example 1: Blog Post Typography

```
article {
  font-size: 16px;
  line-height: 1.6;
}

article h1 {
  text-transform: capitalize;
}
```

```
    text-align: center;
    font-size: 2.5rem;
    margin-bottom: 1rem;
}

article p {
    text-align: justify;
    text-indent: 2em;
    margin-bottom: 1em;
}

article p:first-of-type {
    text-indent: 0; /* No indent for first */
}

article blockquote {
    text-align: center;
    text-transform: capitalize;
    font-style: italic;
    color: #666;
    margin: 2rem 0;
}

article code {
    background-color: #f4f4f4;
    padding: 2px 4px;
    font-family: 'Courier New', monospace;
}

article a {
    text-decoration: underline wavy #3498db;
}
```

9.2 Example 2: E-commerce Product Card

```
.product-card {
    padding: 20px;
}

.product-name {
    font-size: 1.2rem;
    font-weight: 700;
    text-transform: capitalize;
    margin-bottom: 0.5rem;
}

.price {
    font-size: 1.5rem;
    color: #e11d48;
}

.old-price {
    text-decoration: line-through;
```

```
    color: #999;
    font-size: 0.9rem;
}

.description {
    color: #666;
    text-align: left;
    line-height: 1.5;
    max-height: 3em;
    overflow: hidden;
    text-overflow: ellipsis;
    display: -webkit-box;
    -webkit-line-clamp: 3;
    -webkit-box-orient: vertical;
}

.badge {
    text-transform: uppercase;
    font-weight: 600;
    font-size: 0.75rem;
    letter-spacing: 1px;
    padding: 4px 8px;
    background-color: #f0f0f0;
}
```

9.3 Example 3: Navigation Link Styling

```
nav a {
    text-decoration: none;
    text-transform: uppercase;
    font-weight: 600;
    letter-spacing: 1px;
    color: #333;
    padding: 10px 15px;
    display: inline-block;
    transition: all 0.3s ease;
}

nav a:hover {
    color: #3498db;
    text-decoration: underline solid 2px;
}

nav a:active {
    text-decoration: underline solid 3px;
}

nav a.active {
    text-decoration: underline solid #3498db 3px;
    text-decoration-skip-ink: none;
}
```

9.4 Example 4: Heading with Effects

```
h1 {
  font-size: 3rem;
  font-weight: 800;
  text-transform: uppercase;
  letter-spacing: -2px;
  text-shadow: 2px 2px 4px rgba(0, 0, 0, 0.1);
  text-align: center;
  margin-bottom: 1rem;
}

h1.glowing {
  color: #fff;
  text-shadow:
    0 0 10px #3498db,
    0 0 20px #2980b9,
    0 0 30px #0066cc;
  background: linear-gradient(135deg, #1a1a1a, #333);
  padding: 20px;
  border-radius: 8px;
}

h1.outlined {
  color: #fff;
  text-shadow:
    -2px -2px 0 #333,
    2px -2px 0 #333,
    -2px 2px 0 #333,
    2px 2px 0 #333,
    0 0 10px rgba(0, 0, 0, 0.5);
}
```

9.5 Example 5: Truncated List Items

```
ul.list-items {
  list-style: none;
  padding: 0;
}

ul.list-items li {
  padding: 10px;
  margin: 5px 0;
  background-color: #f9f9f9;
  border-left: 4px solid #3498db;
}

ul.list-items .title {
  font-weight: 600;
  max-width: 300px;
  white-space: nowrap;
  overflow: hidden;
  text-overflow: ellipsis;
}
```

```

display: block;
}

ul.list-items .description {
  font-size: 0.9rem;
  color: #666;
  margin-top: 5px;
  display: -webkit-box;
  -webkit-line-clamp: 2;
  -webkit-box-orient: vertical;
  overflow: hidden;
  text-overflow: ellipsis;
}

```

10 Browser Support

10.1 Text Formatting Properties Support

Property	Chrome	Firefox	Safari	Edge
text-align	All	All	All	All
text-decoration	All	All	All	All
text-transform	All	All	All	All
text-shadow	All	3.5+	1.1+	All
text-overflow	4+	7+	3.1+	All
white-space	All	All	All	All
word-break	All	15+	All	All
-webkit-line-clamp	All	68+	All	All
hyphens	All	43+	5.1+	All

11 Accessibility Considerations

11.1 Text Formatting and Accessibility

- **text-transform:** Only visual; doesn't change actual text
- **text-decoration:** Ensure underlines are clear
- **text-shadow:** Don't reduce contrast or readability
- **Truncation:** Provide full text in tooltip or expand
- **text-align:** Use left for body text (better readability)
- **Color:** Ensure sufficient contrast ratios

11.2 Accessible Text Truncation

```

.truncated {
  width: 200px;
  white-space: nowrap;
  overflow: hidden;
}

```



```

text-overflow: ellipsis;
/* Provide full text to screen readers */
title: "Full text here";
}

/* Or with aria-label */
<span class="truncated" aria-label="Full descriptive text">
  Truncated text...
</span>

```

12 Common Mistakes

12.1 Mistake 1: Misusing text-transform

Bad:

```

/* Changes only appearance */
h1 {
  text-transform: uppercase;
}

/* But data still stores "Hello World" */

```

Good:

```

/* If uppercase is needed in data, use JavaScript */
const text = "Hello World";
const uppercase = text.toUpperCase();

/* Or only use text-transform for visual styling */
h1 {
  text-transform: uppercase;
}

/* Ensure actual data is correct */

```

12.2 Mistake 2: Poor Text Truncation

Bad:

```

.truncated {
  max-width: 100px;
  /* Text just breaks and wraps */
}

```

Good:

```

.truncated {
  max-width: 100px;
  white-space: nowrap;
  overflow: hidden;
  text-overflow: ellipsis;
  /* Clean truncation with ellipsis */
}

```

12.3 Mistake 3: Decorative Text Accessibility

Bad:

```
h1 {
  text-shadow: 5px 5px 10px #000;
  /* Text becomes hard to read */
  color: #333; /* Low contrast with shadow */
}
```

Good:

```
h1 {
  text-shadow: 1px 1px 2px rgba(0, 0, 0, 0.2);
  color: #fff; /* High contrast */
  /* Subtle shadow enhances readability */
}
```

13 Advanced Techniques

13.1 Multi-line Text Truncation with Fade

```
.fade-text {
  height: 4.5em; /* 3 lines */
  line-height: 1.5;
  overflow: hidden;
  position: relative;
}

.fade-text::after {
  content: '';
  position: absolute;
  bottom: 0;
  left: 0;
  width: 100%;
  height: 1.5em; /* One line height */
  background: linear-gradient(transparent, white);
}
```

13.2 Custom Text Styles with CSS

```
/* Underline animation */
a {
  text-decoration: none;
  border-bottom: 2px solid transparent;
  transition: border-color 0.3s ease;
}

a:hover {
  border-bottom: 2px solid #3498db;
}
```

```

/* Gradient text (WebKit) */
h1 {
  background: linear-gradient(135deg, #667eea, #764ba2);
  -webkit-background-clip: text;
  -webkit-text-fill-color: transparent;
  background-clip: text;
}

```

14 Summary: Key Takeaways

14.1 Common Text Properties

Property	Purpose	Common Values
text-align	Horizontal alignment	left, center, right, justify
text-decoration	Lines on text	underline, overline, line-through
text-transform	Change case	uppercase, lowercase, capitalize
text-shadow	Shadow effect	2px 2px 4px rgba(...)
text-overflow	Overflow behavior	ellipsis, clip
white-space	Space handling	normal, nowrap, pre
letter-spacing	Char spacing	2px, 0.1em

14.2 Best Practices

1. Use text-align: left for body text (LTR)
2. Keep text-shadow subtle (0.5-2px offset)
3. Avoid excessive text-transform usage
4. Ensure truncation provides fallback (tooltip, aria-label)
5. Test text properties across browsers
6. Maintain sufficient color contrast
7. Use semantic HTML (strong, em) with CSS
8. Consider accessibility when styling text
9. Document unusual text formatting
10. Test on different screen sizes

14.3 Quick Reference

```

/* Professional heading */
h1 {
  text-align: center;
  text-transform: capitalize;
  letter-spacing: 1px;
  text-shadow: 1px 1px 2px rgba(0, 0, 0, 0.1);
}

```

```
/* Truncated text */
.truncated {
  white-space: nowrap;
  overflow: hidden;
  text-overflow: ellipsis;
}

/* Multi-line truncation */
.truncated-lines {
  display: -webkit-box;
  -webkit-line-clamp: 3;
  -webkit-box-orient: vertical;
  overflow: hidden;
}

/* Justified text with hyphens */
article {
  text-align: justify;
  hyphens: auto;
  lang: en;
}

/* Link styling */
a {
  text-decoration: underline;
  text-decoration-color: blue;
  text-decoration-skip-ink: auto;
}
```

15 Conclusion

CSS text formatting properties are essential tools for creating professional, readable, and visually appealing web content. Understanding how to effectively use these properties transforms plain text into engaging, well-structured content.

Key principles to remember:

1. Text alignment affects readability
2. Text decoration should enhance, not distract
3. Transform properties are visual only
4. Proper spacing improves comprehension
5. Shadows should be subtle
6. Truncation needs fallback content
7. Always consider accessibility
8. Test across browsers and devices

9. Use semantic HTML first, CSS second
10. Maintain consistent styling

With mastery of text formatting properties, you'll create web content that is both beautiful and accessible, providing excellent user experiences across all devices and contexts.

Happy coding!